

Planning ahead

Writing software is a trade-off between doing things the right way and getting things done in time. Several design techniques exist to help strike the right balance.

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The three big pitfalls

Kids use plastic construction blocks to create an infinite variety of objects, such as castles, boats, and spaceships. A block used in an aeroplane can later be used in a car. Everything is flexible, versatile, and multi-purpose. Keeping these qualities in mind can help to avoid common pitfalls that may pop up in the future.

IN DEPTH

Spaghetti code

This is a term for code that looks and reads like spaghetti: a tangled, twisted mess. Have you ever tried to pull a single noodle out of a jumble of spaghetti, only to have it break? Or, as you carefully removed the noodle, realized that it was way longer than expected? Neither of these are experiences you want when debugging code.

1. Scaling

Websites and apps progressively get more traffic. The software and the servers should be designed to handle a future increase in users, or at least make it easy to upgrade.

2. New features

Companies tweak their products all the time. Good code should be flexible enough for developers to make changes without wrecking the entire program.

3. New developers

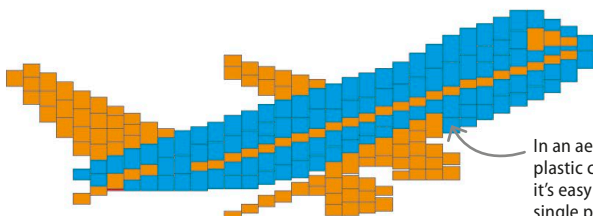
People might move, retire, or get promoted. A single project has developers going in and out, so code should be written in the clearest way possible so that people unfamiliar with it can understand it with ease.

Design patterns

In computer science, a design pattern is a generally repeatable solution to a commonly occurring problem. It can be seen as a strategy that makes code more like building blocks. There's a pattern for every tricky situation, from creating objects to coordinating messages and protecting data. Each pattern isolates specific behaviours and assigns them to particular objects – or files, functions, and chunks of code – in order to make code as flexible and versatile as possible.

▽ Changing parts

To modify the wings of a plane made of plastic construction blocks, the entire aeroplane doesn't have to be destroyed. All that's required is to snap the wings off, build new wings, and snap them on. Design patterns in code creation help to make this process as seamless as possible.



In an aeroplane made of plastic construction blocks, it's easy to swap out a single piece for another.

Code written in an inflexible manner is like a real-life plane, where everything is permanently soldered together.

